GREY HUNTER

SYSTEM ARCHITECTURE & MULTI-PHASE IMPLEMENTATION PLAN

Autonomous Mobile Security Auditing & Defensive Penetration Testing Platform
Version 2.0.0 • Release Roadmap • GitHub Production Specification

1. Executive Summary & Mission Charter

Grey Hunter is an enterprise-grade mobile cybersecurity auditing, network discovery, and ethical testing companion. Built to bridge command-line auditing frameworks and modern mobile field operations, it empowers certified security professionals, penetration testers, and purple teams to assess infrastructure security from a mobile device while maintaining cryptographic verification, authorized scoping, and defense-first principles.

Core Architectural Pillars:

- **Zero-Trust Identity:** Hardware-backed authentication utilizing Android Keystore, StrongBox, and biometric verification.
- **Isolated Execution:** Native Linux security binaries running inside sandboxed environments with strict privilege dropping.
- **Automated Vulnerability Telemetry:** Dynamic CVSS v3.1 scoring, CVE correlation, and immediate remediation playbooks.
- **Encrypted Offline Storage:** SQLCipher 256-bit AES database ensuring zero data exposure even on compromised mobile devices.

2. System Architecture & Component Topology

The platform utilizes a multi-tiered architecture that separates the Presentation Layer (React Native / Mythic Dragon UI) from the Orchestration Engine, Hardware Bridges, and Local/Cloud Execution runtimes.

Layer	Primary Technologies	Core Responsibilities & Security Controls
Presentation	React Native 0.73, TypeScript 5.3, Redux Toolkit, Mythic Dragon UI	Cyberpunk terminal styling, real-time telemetry rendering, role selection, responsive tablet/mobile layout.
Hardware Bridge	Android Keystore, BiometricPrompt, PTY Native Bridge	Secure key derivation, master credential wrapping, pseudo-terminal session multiplexing.
Execution Sandbox	Termux Runtime, PRoot / Chroot, Raw Sockets (CAP_NET_RAW)	Native binary invocation (Nmap, Netdiscover, Masscan), command sanitization, and output piping.
Telemetry & Scoring	XML/JSON Streaming Parsers, CVSS 3.1 Calculator, CVE Correlator	Transform raw terminal stream into structured vulnerability cards, risk scores, and mitigation advice.
Encrypted Storage	SQLCipher, AES-256-GCM, EncryptedSharedPreferences	Zero-knowledge local database holding scan reports, target CIDRs, and cryptographic audit logs.
Cloud Bridge	Node.js / Express, Prisma ORM, PostgreSQL, Socket.io, Docker	Multi-operator team synchronization, encrypted report backups, and continuous intelligence feeds.

3. Multi-Phase Implementation Roadmap

Development is organized into six progressive, rigorous engineering phases to ensure enterprise readiness and zero-compromise security.

Phase	Focus Area	Deliverables & Key Milestones	Security & Verification
Phase 1	**Cryptographic Identity & Auth**	Dragon RegisterScreen, LoginScreen with clearance levels, Android Keystore integration, biometrics.	AES-256-GCM, zero-knowledge password hashing, strict client input sanitization.
Phase 2	**Tool Sandboxing & Execution**	8 new security tools (Amass, Nuclei, Netdiscover, WPScan, RustScan), JSON definitions, PTY terminal bridge.	Command parameter injection prevention, capability checks (CAP_NET_RAW), out-of-scope target blocking.
Phase 3	**Analysis & CVSS Telemetry**	Multi-tab ScanResultsScreen, posture scores (A-F), vulnerability severity matrix, remediation playbooks.	Standardized CVSS v3.1 calculation, sensitive IP data redaction, report sharing verification.
Phase 4	**Encrypted Offline Storage**	SQLCipher encrypted database, offline sync queue, secure PCAP capture storage, DoD wipe routines.	256-bit database encryption at rest, PBKDF2 64,000 rounds, zero leak to external storage.
Phase 5	**Collaborative Teaming**	Multi-operator WebSocket mesh, live session multiplexing, Kanban task boards, encrypted audit logs.	mTLS certificate pinning, end-to-end payload signing, immutable cryptographic engagement logs.
Phase 6	**Hardening & Release**	ProGuard/R8 code obfuscation, anti-tamper runtime hooks, CI/CD GitHub Actions, F-Droid & Play distribution.	OWASP MASVS Level 2 compliance, root/jailbreak detection, automated dependency vulnerability audits.

4. Comprehensive Security Tool Arsenal Matrix

Grey Hunter features 48 integrated defensive and auditing tools spanning 6 distinct cybersecurity domains:

Domain	Key Tools Included	Operational Capabilities
Network Recon & Discovery	Nmap, Masscan, RustScan, Netdiscover, Netscan, ARP-Scan, Hping3, Traceroute	Active/passive ARP discovery, high-speed port scanning (65k ports/3s), OS fingerprinting, ICMP sweeps.
Web App Auditing	Nuclei, WPScan, SQLMap, Nikto, Gobuster, Dirb, WhatWeb	YAML-template vulnerability scanning, WordPress CMS auditing, automated SQLi validation, directory fuzzing.
OSINT & Surface Mapping	OWASP Amass, Subfinder, theHarvester, Recon-ng, SpiderFoot	Passive subdomain discovery, certificate transparency analysis, external attack surface graphing.
Secret & Credential Audit	TruffleHog, Hash-Identifier, Hydra, John the Ripper	High-entropy secret detection, live API key verification in Git, hash classification, credential strength audits.

Wireless & Packet Telemetry	Aircrack-ng, Bettercap, Tcpdump, Wireshark, Tshark	802.11 monitor mode inspection, PCAP capture and analysis, rogue access point identification.
Crypto & Secure Comms	OpenSSL, GPG, Netcat, Socat, SSH Client	TLS/SSL cipher suite validation, OpenPGP message signing, encrypted tunnels, raw network pipe debugging.

5. Ethical Framework & Safety Compliance

1. Pre-Authorized Scope Enforcement: All scan operations require target CIDR or domain validation. Unscoped IPs are blocked by application policy.
2. Defensive Orientation: Tools perform non-destructive reconnaissance and vulnerability auditing. No automated exploitation or malicious payloads are included.
3. Cryptographic Audit Trail: Every tool execution generates an immutable log entry signed with the operator's private key containing target, timestamp, and parameters.
4. Legal Compliance: Operators must sign the Grey Hunter Rules of Engagement oath during registration before tool capabilities are unlocked.

Document Status: APPROVED FOR GITHUB REPOSITORY DEPLOYMENT • Grey Hunter Core Engineering Team